$$L(y_i, c) = (y_i - c)^2$$

$$g(c) = \underset{c}{\text{minimize}} \sum_{i=1}^{n} (y_i - c)^2$$

$$g'(c) = 2 \sum_{i=1}^{n} (y_i - c)^1 (-1) \equiv 0$$

solve for c

$$\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} c = 0$$

$$nc$$

$$\sum_{i=1}^{n} y_i = nc$$

$$c = \frac{\sum_{i=1}^{n} y_i}{n} = \bar{y}$$

SLR

- as least squares

$$\min_{\beta_0, \beta_1} h(\beta_0, \beta_1) = \sum_{i=1}^{n} \left(y_i - \underbrace{(\beta_0 + \beta_1 x_i)}_{\text{prediction}}\right)^2$$

$$\frac{\partial h}{\partial \beta_0} = 2 \sum_{i=1}^{n} \left(y_i - (\beta_0 + \beta_1 x_i)\right)^1 (-1) \equiv 0$$ — solve for β_0 plug into 2nd eq.

$$\frac{\partial h}{\partial \beta_1} = 2 \sum_{i=1}^{n} \left(y_i - (\beta_0 + \beta_1 x_i)\right)^1 (-x_i) \equiv 0$$

$$\sum_{i=1}^{n} y_i - n\beta_0 - \beta_1 \sum_{i=1}^{n} x_i = 0$$

$$\beta_0 = \frac{\sum_{i=1}^{n} y_i - \beta_1 \sum x_i}{n} = \bar{y} - \beta_1 \bar{x}$$

plug into 2nd eq + solve for β_1

$$\sum_{i=1}^{n} y_i x_i - \beta_0 \sum_{i=1}^{n} x_i - \beta_1 \sum_{i=1}^{n} x_i^2 = 0$$

$$\sum y_i x_i - (\bar{y} - \beta_1 \bar{x}) \sum x_i - \beta_1 \sum x_i^2 = 0$$

$$\sum y_i x_i - \bar{y} \sum x_i + \beta_1 \bar{x} \sum x_i - \beta_1 \sum x_i^2 = 0$$

$$\sum y_i x_i - \bar{y} \sum x_i = \beta_1 \sum x_i^2 - \beta_1 \bar{x} \sum x_i$$

$$\frac{\sum y_i x_i - \bar{y} \sum x_i \left(\frac{n}{n}\right)}{\sum x_i^2 - \bar{x} \sum x_i \left(\frac{n}{n}\right)} = \beta_1$$

Note:

$$\sum (x_i - \bar{x})(y_i - \bar{y})$$

$$\sum \left[x_i y_i - x_i \bar{y} - y_i \bar{x} + \bar{x}\bar{y}\right]$$

$$\sum x_i y_i - n\bar{x}\bar{y} \cancel{- n\bar{x}\bar{y}} \cancel{+ n\bar{x}\bar{y}}$$

$$\beta_1 = \frac{\sum y_i x_i - n\bar{y}\bar{x}}{\sum x_i^2 - n(\bar{x})^2} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})(x_i - \bar{x})^2}$$

$$= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \boxed{\frac{S_{xy}}{S_{xx}}}$$

$$\boxed{\hat{\beta}_1 = \frac{S_{xY}}{S_{xx}} \qquad \hat{\beta}_0 = \bar{Y} - \bar{x}\hat{\beta}_1}$$

Least squares estimators of β_1 and β_0

Maximum Likelihood

$Y \sim N(\mu, \sigma^2)$

PDF

given mean + var

$$f_Y(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y-\mu)^2}$$

function of y

$-\infty < y < \infty$
$-\infty < \mu < \infty$
$\sigma > 0$

$$P(Y > \underbrace{\mu + \sigma}_{y}) =$$

Being given a y, what is the most likely μ to have created that y (fix σ^2)

given

$$L(\mu \mid y, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y-\mu)^2}$$

Likelihood $\quad \mu$

Maximize the likelihood as a function of μ, get $\hat{\mu}$

Random Sample of Y's

$\rightarrow Y_1 \overset{iid}{\sim} N(\mu, \sigma^2)$

$$\left.\begin{array}{l} Y_1 \sim N(\mu, \sigma^2) \\ Y_2 \sim N(\mu, \sigma^2) \\ \vdots \\ Y_n \sim N(\mu, \sigma^2) \end{array}\right\} \text{identically distr. part}$$

$A \perp B \Rightarrow P(A \cap B) = P(A)P(B)$

independent

realizations

$$\underbrace{f_{Y_1, Y_2, \ldots, Y_n}}_{\text{joint PDF of these RVs}}(y_1, y_2, \ldots, y_n) = f_{Y_1}(y_1) f_{Y_2}(y_2) \cdots f_{Y_n}(y_n)$$

by independence

$(x^2)(x^5) = x^7$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_1-\mu)^2} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_2-\mu)^2} \cdots \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_n-\mu)^2}$$

$$= \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2}$$

$$L(\mu, \sigma^2 \mid y_1, \ldots, y_n) = \rightarrow$$

$\ln(a \cdot b) = \ln(a) + \ln(b)$

optimize as a function of μ

take log (natural) and optimize that instead (for ease)

$$\ell(\mu \mid \sigma^2, y_1, \ldots, y_n) = \ln(L(\mu \mid \sigma^2, y_1, \ldots, y_n))$$

$$= \ln(\quad) + \ln(e^{\cdots})$$

$$= c + -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2$$

$$\frac{d\ell}{d\mu} = -\frac{1}{\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^1(-1) \equiv 0 \qquad \text{solve for } \mu$$

$$\sum y_i - \sum_{i=1}^{n}\mu = 0$$

$$\sum y_i = n\mu \qquad \Rightarrow \boxed{\hat{\mu} = \bar{Y}}$$

Relate to SLR

$$Y_i = \underbrace{\beta_0 + \beta_1 \boxed{x_i}}_{\substack{\text{fixed} \\ \text{'mean' part}}} + \underset{\substack{\uparrow \\ \text{randomness}}}{\varepsilon_i} \qquad \varepsilon_i \overset{iid}{\sim} N(0,\sigma^2)$$

$$Y_i \overset{ind}{\sim} N(\beta_0+\beta_1 x_i, \sigma^2)$$

$$L(\beta_0,\beta_1,\sigma^2)(x_i,y_i)) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-(\beta_0+\beta_1 x_i))^2}$$

$$\ell(\beta_0,\beta_1,\sigma^2 \mid (x_i,y_i)) = c - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-(\beta_0+\beta_1 x_i)^2)$$

maximizing this as a function of β_0, β_1 is the same as minimizing the negative of the function

$$\min_{\beta_0,\beta_1} -\ell(\beta_0,\beta_1) = \min_{\beta_0,\beta_1} \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-(\beta_0+\beta_1 x_i))^2$$

Same optimization!

$$\underset{\sim}{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \qquad y = \underset{\substack{\uparrow \\ \text{design} \\ \text{matrix}}}{X}\beta + \varepsilon \qquad \hat{\underset{\sim}{\beta}} = (X^TX)^{-1}X^Ty$$

Binomial Distribution

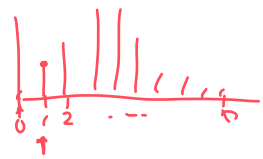
Y = # of successes in a fixed # of trials

$Y \sim Bin(n, p)$

n = # of trials

$p = P(\text{Success})$

$y = 0, 1, 2, \ldots, n$



PMF $\quad Pr(y) = \binom{n}{y} p^y (1-p)^{n-y}$

PMF $\quad$ realization

ex: A sample of data

$y = 7 \quad$ or $\quad y = 13$

use likelihood idea to estimate p given a y

$\ln(ab) = \ln(a) + \ln(b)$

$\ln(a^x) = x \ln(a)$

$$L(p|y) = \binom{n}{y} p^y (1-p)^{n-y}$$

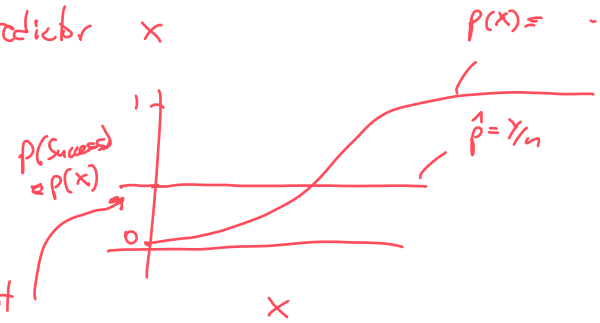
log loss w/ $n = 1$

log likelihood $\quad \ell(p|y) = \ln\left(\binom{n}{y}\right) + \boxed{y \ln(p) + (n-y)\ln(1-p)}$

$\hat{p} = \frac{y}{n}$ $\quad$ sample prop of successes

w/ logistic regression we also have a predictor x

$$p(x) = \frac{e^{-\beta_0 - \beta_1 x}}{1 + e^{-\beta_0 - \beta_1 x}}$$

if $\beta_1 = 0$

$$= \frac{e^{-\beta_0}}{1 + e^{-\beta_0}} = \text{constant}$$